

Closed forms and sharp minima of ωt_c on three classical point-vortex collapse families

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Classical formulas and elementary sharp bounds. Zero confirmed novel findings.

Abstract.

Self-similar collapse of N point vortices is classical. Gröbli (1877) and Novikov–Sedov (1979) found the families; Aref (2010) and Gotoda (2020) wrote the collapse rate A and the spin B as separate functions of a shape angle. Their dimensionless product $\omega_0 t_c = -\frac{B}{2A}$ is the pitch of the logarithmic spiral the configuration traces as it shrinks. This note records three closed forms of that product, with elementary sharp minima. The first expression explicitly specializes Gröbli’s 1877 spiral coefficient; historical priority of the optimized minima remains unconfirmed:

- Three-vortex collapse bound, on Gröbli’s $\Gamma = (1, 1, -\frac{1}{2})$ family: $\omega_0 t_c = \frac{2 - \cos^2 \theta}{\sin(2\theta)} \geq \sqrt{2}$, equality at $\tan \theta = \frac{1}{\sqrt{2}}$.
- The parallelogram lock, on the Novikov–Sedov four-vortex parallelogram: $\omega_0 t_c = \left(\frac{\sqrt{3}}{4}\right) \frac{4 - \cos 2\theta}{\sin(2\theta)} \geq 3\frac{\sqrt{5}}{4}$, equality at $\cos 2\theta = \frac{1}{4}$.
- The quincunx lock, on the Novikov–Sedov five-vortex quincunx: $\omega_0 t_c = \left(\frac{3}{16}\right) \frac{7 - 4 \cos 2\theta}{\sin(2\theta)} \geq 3\frac{\sqrt{33}}{16}$, equality at $\cos 2\theta = \frac{4}{7}$.

These are descriptive project labels. Cite the classical sources; no ownership or first-discovery claim follows from this note.

1. What is claimed, and what is not

Let $z_{m(t)} \in \mathbb{C}$ be the positions of N point vortices of circulations Γ_m , evolved by the 2π -periodic Biot–Savart law. A configuration is self-similar when there exist real A, B , independent of m , with

$$A + iB = \frac{i}{2\pi z_m} \sum_{n \neq m} \frac{\Gamma_n}{\bar{z}_m - \bar{z}_n}. \quad (1)$$

If $A < 0$, the vortices collide at the origin at the finite time $t_c = -\frac{1}{2A}$, rotating with instantaneous angular velocity ω so that $\omega_0 t_c = -\frac{B}{2A}$ is scale-invariant [1], [2], [3].

Mathematical scope. The displayed formulas and elementary minima are checked on their stated classical families. This is a derivation and implementation record, not a claim that these results were first discovered here.

Not claimed. The motion (Gröbli; Novikov–Sedov). The product as the pitch of a log spiral (Aref). The functions $A(\theta)$, $B(\theta)$ (Gotoda). The fastest-collapse time $t_c = 4\frac{\pi}{3}$ on the three-vortex family at unit separation (Leoncini, Kuznetsov and Zaslavsky). Any published isolated lock listed in the search ledger of GENChase.

2. Three-vortex collapse bound

Family. $\Gamma = (1, 1, -\frac{1}{2})$, positions $z_1 = 0$, $z_2 = 1$, $z_3 = \frac{1}{2} + \left(\frac{\sqrt{3}}{2}\right)e^{i\theta}$ with $0 < \theta < \frac{\pi}{2}$. These lie on the angular-impulse circle $L = 0$ and collapse self-similarly [4]. The parameter θ is the third vortex's place on that circle, not an interior angle of the triangle.

$$\omega_0 t_c = \frac{2 - \cos^2 \theta}{\sin(2\theta)} \geq \sqrt{2}.$$

Equality uniquely on this arc at $\tan \theta = \frac{1}{\sqrt{2}}$, a triangle of interior angles 22.5° , 45° , 112.5° .

Proof of the minimum. Set $u = \tan \theta > 0$. Then $\omega_0 t_c = u + \frac{1}{2u}$ and

$$\omega_0 t_c - \sqrt{2} = \frac{(\sqrt{2}u - 1)^2}{2u} \geq 0.$$

At $\theta = 45^\circ$ the product is $\frac{3}{2}$. The same product splits, at this length, as

$$t_c = \left(\frac{\pi}{3}\right)\left(4u + \frac{1}{u}\right), \quad 2\pi\omega_0 = \frac{3(2u^2 + 1)}{4u^2 + 1},$$

which are Aref's τ and Ω in this family's angle [2]. Their product does not depend on the length unit. The collapse time itself has a unique minimum $4\frac{\pi}{3}$ at $u = \frac{1}{2}$; that number is already in Leoncini, Kuznetsov and Zaslavsky [5] and is not a second identity.

Label. Three-vortex collapse bound. This is a descriptive project label for a specialization of Gröbli's published formula.

3. Parallelogram lock

Family. $\Gamma = (1, 1, -2 - \sqrt{3}, -2 + \sqrt{3})$ at the vertices of a parallelogram whose diagonals meet at the origin, diagonal ratio $\frac{d_1}{d_2} = \sqrt{2 + \sqrt{3}}$, angle $\theta \in (0, \frac{\pi}{2})$ between the diagonals. These have $L = 0$ and collapse self-similarly [6]. The reciprocal pair $\Gamma = (1, 1, -2 + \sqrt{3}, -2 - \sqrt{3})$ with the diagonals swapped carries the same product.

$$\omega_0 t_c = \left(\frac{\sqrt{3}}{4}\right) \frac{4 - \cos 2\theta}{\sin(2\theta)} \geq \frac{3\sqrt{5}}{4}.$$

Equality uniquely on this arc at $\cos 2\theta = \frac{1}{4}$.

Proof of the minimum. Set $\varphi = 2\theta \in (0, \pi)$. The quotient $\frac{4 - \cos \varphi}{\sin \varphi}$ has a unique critical point on $(0, \pi)$ at $\cos \varphi = \frac{1}{4}$, where $\sin \varphi = \frac{\sqrt{15}}{4}$ and the quotient equals $\sqrt{15}$. Multiplying by $\frac{\sqrt{3}}{4}$ yields $3\frac{\sqrt{5}}{4}$. The sign of the first derivative on either side shows a minimum. Endpoints send the product to infinity. At $\theta = 45^\circ$ the product is $\sqrt{3}$. The product is $-\frac{B}{2A}$ from Gotoda's eq. (3.13) [3] specialized to this parallelogram; direct Biot–Savart on the 2π kernel matches that closed form to machine precision.

Label. Parallelogram lock, named for the geometry. Its formula follows established Novikov–Sedov and Gotoda rates.

4. Quincunx lock

Family. $\Gamma = (-1, -1, \frac{1}{2}, \frac{1}{2}, -\frac{3}{4})$, four vortices at the vertices of a parallelogram and one at the crossing of the diagonals, diagonal ratio $\frac{d_1}{d_2} = \frac{1}{\sqrt{2}}$, angle $\theta \in (0, \frac{\pi}{2})$ between the diagonals. These have $L = 0$ and collapse self-similarly [3], [6].

$$\omega_0 t_c = \left(\frac{3}{16}\right) \frac{7 - 4 \cos 2\theta}{\sin(2\theta)} \geq \frac{3\sqrt{33}}{16}.$$

Equality uniquely on this arc at $\cos 2\theta = \frac{4}{7}$.

Proof of the minimum. Set $\varphi = 2\theta \in (0, \pi)$. The quotient $\frac{7-4\cos\varphi}{\sin\varphi}\varphi$ has a unique critical point on $(0, \pi)$ at $\cos\varphi = \frac{4}{7}$, where $\sin\varphi = \frac{\sqrt{33}}{7}$ and the quotient equals $\sqrt{33}$. Multiplying by $\frac{3}{16}$ yields $3\frac{\sqrt{33}}{16}$. Endpoints send the product to infinity. At $\theta = 45^\circ$ the product is $\frac{21}{16}$. The product is $-\frac{B}{2A}$ from Gotoda's five-vortex $A(\theta)$, $B(\theta)$ ($\gamma_3 \neq 0$); direct Biot–Savart matches. Gotoda plots this family as Hamiltonian against collapse rate and does not form the product or state its minimum [3], [7].

Label. Quincunx lock, named for the geometry. Its formula follows the published five-vortex family; priority of the elementary minimum is unconfirmed.

5. Attribution and historical limits

The first expression is a specialization and reparameterization of Gröbli's §10 spiral coefficient. Set $a = \frac{\sqrt{3}}{\cos}\theta$ in

$$P = \frac{2a^2 - 3}{2\sqrt{3a^2 - 9}}$$

to recover the stated product. The original scan uses $\mu_1\mu_2\mu_3$ in the relevant denominator; the repository follow-up documents the transcription check [4].

The other two expressions follow published Novikov–Sedov/Gotoda rate formulas. Their optimization is elementary: for $a > |b|$, $c > 0$ and $0 < \varphi < \pi$,

$$\frac{c(a - b \cos \varphi)}{\sin \varphi} \geq c\sqrt{a^2 - b^2}, \quad \cos \varphi_* = \frac{b}{a}.$$

The square identity $(a - b \cos \varphi)^2 - (a^2 - b^2) \sin^2 \varphi = (a \cos \varphi - b)^2$ verifies the bound.

No earlier exact minima were located in the inspected passages, but that absence does not establish historical originality or substantial novelty. The complete older two-ring literature remains an access gap for related polygon candidates. The project currently has zero confirmed novel findings.

[Read the original-source comparison](#) and [the novelty audit](#) before making a priority claim. No specialist endorsement is implied.

6. Reuse and file provenance

Anyone may use the mathematics. Cite the classical sources for their formulas and cite GENChase when using its exposition or implementation. Repository dates record this project's work; they

do not establish first discovery. The corrected text, source and PDF have updated SHA-256 fingerprints in `identities/HASHES.txt`.

This version replaces the prior note’s unsupported naming and priority language at the author’s request. The derivations remain available under descriptive labels.

References

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